Grammar

```
e ::= n | true | false
  | e1 + e2
  | e1 <= e2
  | if e1 then e2 else e3
```

key

$$\mathbb{B} \equiv \text{Bool} \mid \mathbb{Z} \equiv \text{Int} \mid \tau \equiv \text{Generic}$$

If Lang

Axioms:

$$\frac{}{\Gamma \vdash \text{n} : \mathbb{Z}}\ \text{T-Int} \qquad \frac{}{\Gamma \vdash \text{T} : \mathbb{B}}\ \text{T-True} \qquad \frac{}{\Gamma \vdash \text{F} : \mathbb{B}}\ \text{T-False}$$

inference rules:

$$\frac{\vdash e_1 \colon \mathbb{Z} \quad \vdash e_2 : \mathbb{Z}}{\Gamma \vdash e_1 + e_2 \colon \mathbb{Z}}\ \text{T-Add} \qquad \frac{\vdash e_1 \colon \mathbb{Z} \quad \vdash e_2 \colon \mathbb{Z}}{\vdash e_1 \leq e_2 \colon \mathbb{B}}\ \text{T-leq} \qquad \frac{\vdash e_1 \colon \mathbb{B} \quad \vdash e_2 \colon \tau \quad \vdash e_3 \colon \tau}{\vdash \text{if } e_1 \text{ then } e_2 \text{ else } e_3 \colon \tau}\ \text{T-if}$$

some rules that make you questions your life

$$\frac{\vdash e_1 \colon \tau \qquad \vdash e_2 \colon \tau'}{\vdash \text{if true then } e_1 \text{ else } e_2 \colon \tau}\ \text{if-true}$$

Grammar

```
e ::= n | true | false
  | e1 + e2
  | e1 <= e2
  | if e1 then e2 else e3
```